

How a simple increase in the number of items can enhance the reliability of judgment data: The case of island experiments

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Abstract: Replication is an important aspect of experimental research and it is therefore crucial that collected datasets are reliable. *Reliability* refers to the precision of measurement and thus informs the replicability of experiments: more precise measurements are more dependable for future reference. Formally defined as the ratio of true score variance to the total variance, reliability can be achieved by fine-tuning the instrument of measurement or by collecting a sufficiently large number of observations, as statistical models operating on larger sample sizes are in a better position to estimate the true score associated with a population. The present paper uses Generalizability Theory to explore the variation patterns in 52 distinct datasets from studies that used comparable experimental designs to investigate different types of island effects in different languages. Effect sizes (DD-scores) are commonly reported and used for comparative purposes in discussions on island effects. The present paper argues that caution is warranted when island effect sizes are compared: the analyses reveal that the reliability of datasets from island experiments is only moderate. At the same time, the analyses show that reliability can be improved relatively simply, namely, by increasing the number of items tested in the experiment to six per condition.

Keywords: replication; reliability; judgment data; experimental syntax; island effects

1. Introduction

Replication of experimental work is integral to the scientific method. Yet, for many different reasons, it is not guaranteed that published experimental research can always be replicated. This phenomenon, dubbed the ‘replication crisis’, has received much recent attention in the social sciences (Shrout & Rodgers, 2018). The field of (psycho)linguistics has not been able to escape from concerns of replication failure either (Sönning & Werner, 2021; Dempsey & Christianson, 2022): as the object of study is an ‘inextricably social phenomenon’, Grieve (2021: 1344) argues that variation is often due to unknown or unidentified factors. This type of variation impedes the reliability of datasets, even when experiments are carefully designed. That is, language researchers can take action and e.g., preregister their experiments to reduce biases (Roettger, 2021), collect data according to established methods (Gibson & Fedorenko, 2013; but see Sprouse & Almeida, 2013), or seek other ways to reduce noise (Christianson et al., 2022), but the nature of the research object will inevitably trouble the waters: unexplained or unknown variation is inherent to linguistic data and will always reduce the reliability of a dataset. Vasishth and Gelman (2018) discuss various ways for language researchers to deal with variation in (psycho)linguistic data, in order for them to nuance their (typically overly strong) claims based on less than reliable datasets. They show that sorting out the variation patterns in experimental datasets is an important component of the statistical analysis, even regardless of the outcome of significance tests. For one thing, the variation patterns can be indicative of the replication potential of any outcome.

More specifically, reproduction of an experiment (under comparable circumstances) ideally yields results similar to those from the original experiment.¹ Researchers wish for their data to be *reliable*. The qualification ‘ideally’ is added here, because consistency over experiments is by no means a given: as alluded to above, countless factors, many of which unavoidable, may somehow influence the measurement procedure and contribute ‘error’ to the scores. This is at issue in the social sciences especially, given that fallibility is inherent to human measurement. But since precise measurements are essential to scientific endeavors, one of the researchers’ primary tasks should be to make an effort to minimize measurement error. One way to improve the precision of measurement is by changing the tools we have in our shed: improving the apparatus and ‘asking the right questions in the right way’. Indeed, there is a growing literature on research methodology in experimental syntax (e.g., Schütze, 1996/2016; Cowart, 1997; see Sprouse & Almeida, 2017; Langsford et al., 2018; Marty et al., 2020; Schoenmakers, 2023, for more recent contributions). Another way to improve the precision of measurement is by increasing the sample size. That is, in sorting out their data, language researchers take average scores over many measurements (e.g., when a linguist wants to generalize beyond a single sentence or the judgments of a single person), on the assumption that the measurement errors will cancel each other out when the dataset is large enough. If, for instance, you want to guess the average speech rate of women from the state of Utah, you are in a better position to do so when you collect speech data from 1.000 women instead of from only one person. The question, then, is: how many instances are required for an acceptably precise measurement? Or, more specifically for the purposes of experimental syntacticians: how many participants should we ask to rate how many sentences? The answer to this question depends on the reliability of the dataset (i.e., the extent to which the collected ratings are correlated with each other) and on the specific sources of measurement error. If, for instance, prior research has shown that participants are in perfect agreement about the acceptability of a set of sentences, but there is considerable variability in the acceptability scores of the individual sentences, it is sensible to increase the number of sentences (or vice versa). Note, however, that increasing the sample size only improves measurement precision up to a certain point.

Social scientists often perform statistical power analyses to determine the number of participants required to run a well-powered experiment (Brysbaert, 2019). They wish to run properly powered experiments, but at the same time, testing more participants than is strictly necessary can be considered a waste of time and resources. Statistical power is the probability that an effect that exists in the real world is observed in an experimental study (see also Sprouse & Almeida, 2017), and is computed based on a fixed effect size,² a set significance level (or alpha), and a chosen number of participants. Keeping the effect size and the significance level constant, a researcher can calculate the number of participants required for an experiment with, say, 80% power. A somewhat less common alternative to improve the statistical power is to increase the number of observations per participant (see Rouder & Haaf, 2018). As noted, the effectiveness of this choice depends on the variability and reliability of the variable: in most cognitive studies, participants are likely to vary to a larger extent than experimental trials. Still, as Brysbaert (2019: 19) puts it, “[m]ost cognitive researchers have an intuitive understanding of the requirement for reliable measurements, because they rarely rely on a single observation per participant per condition.” It goes to show that reliability and sample size are important ingredients of a proper power analysis.³

^{1.} It is worth noting that more journals in linguistics have started publishing open access articles and have adopted policies for researchers to publish their stimuli, data, and code online, following the FAIR principles (Findable, Accessible, Interoperable, Reusable) to stimulate reproducibility of existing studies and replication of their findings. Such policies enhance transparency and are inductive of reproducibility, as other researchers can reexamine, reanalyze, or reuse these data (Laurinavichyute et al., 2022).

^{2.} Brysbaert (2019) discusses various problems involved in fixing the effect size for this computation: there are usually only a handful of sufficiently similar studies, and the effect sizes reported in published studies are likely overestimated.

^{3.} Researchers are encouraged to report the reliability coefficients of their experiments (see also Gelman, 2018), as it would allow other researchers to compare the reliability associated with particular instruments and test features so that they can make more substantiated choices for their own experiments.

This paper explores the impact of an increase in the number of items on the reliability of the datasets in experimental syntactic research. To that end, it reports on a series of reliability analyses of 52 datasets from experiments similar in design to test island effects in various languages and configurations. The common operationalization of island experiments is discussed in Section 3, after a brief background on statistical reliability is presented in Section 2. In particular, Generalizability Theory (Cronbach et al., 1972; Shavelson & Webb, 1991; Brennan, 2001) is used to determine the reliability of the selected datasets. Their average reliability turns out to be only moderate, with few of the datasets exhibiting what is considered good reliability. This is problematic in light of the fact that researchers tend to report and compare the island effect sizes (in terms of the *differences-in-differences scores*; see e.g., Sprouse & Villata, 2021: §9.4): such comparisons are risky business when the island effects are not sufficiently reliable. It is thus important to find ways to improve the reliability of these datasets. A crucial observation with regard to the selected experiments is that most of them only tested two items per condition. An advantage of ‘G-Theory’ is that it can perform *decision studies*, or ‘D-studies’, in which the variance components of a given dataset (which are collected in *generalizability studies*, or ‘G-studies’) are used to estimate a design in which error is minimized (Rajaratnam, 1960; Cronbach et al., 1972; Shavelson & Webb, 1991: Chapter 6). D-studies can thus be used to optimize measurement through determining the number of items required to achieve good reliability. Results from a series of D-studies performed on the selected datasets indicate that an increase of the number of items in experimental explorations of island phenomena to *six per condition* (i.e., 24 items in a 2×2 factorial design, using a Latin Square distribution) would considerably improve their reliability.

2. Statistical reliability

Variability is inherent to human measurement. At the participant level, it can be caused by fatigue, learning effects, diverging uses of the scale, or idiosyncratic judgment criteria (Rietveld & van Hout, 1993). Peer et al. (2021) show that participants recruited on various online platforms are not always attentive, motivated, do not always seem to understand the instructions, or are sometimes otherwise disengaged (see also Christianson et al., 2022, for discussion on participant motivation). As Dahl (1979) wittily suggests, however, variation between participants may also be caused by seemingly arbitrary factors such as what the participant had for breakfast. At the item level, variation may be due to the combinatorial semantics or pragmatic appropriateness of the stimulus sentences, the use of particularly demanding, ill-suited, unfamiliar, infrequent, lengthy, or archaic lexical items and/or syntactic constructions, but perhaps variation between items may also be due to arbitrary factors such as the current phase of the moon – that is to say, we can safely assume that there will always be unidentified or random variance in judgment data due to the nature of experimental research in (psycho)linguistics (see also Häussler & Juzek, 2021). Put differently, perfect measurement simply does not exist in experimental research with human instruments. Therefore, the rationale in experimental research is that, at least for experiments with a large enough sample size, the residual errors will cancel each other out when the data are averaged: the mean score will then be in a better position to accurately represent the ‘true’ (acceptability) score (Rietveld & van Hout, 1993).⁴

Classical test theory assumes that observed scores are composed of two distinct components: a true score and an error component. The true score component represents the ‘real’ acceptability of a target sentence, but the observed score will always contain some error. The goal of classical test theory is to estimate the true score as accurately as possible, on the basis of the average scores of a hypothetically infinite number of measurements. The internal consistency, or reliability, can then be calculated by e.g.,

⁴ Nowadays, experimental linguists mostly use linear mixed effects regression models to test their hypotheses (see Winter, 2019, for an introduction). These models are ‘mixed’, because they include not only the ‘fixed’ factors that were manipulated in the experiment, but also ‘random’ effects which take into account the fact that the collected data points are not fully independent from each other: there are multiple datapoints for each item (sentence) from more than a single participant.

the intraclass correlation coefficient (ICC; Shrout & Fleiss, 1979), which represents the ratio of the true score to the total variance, and is obtained through ANOVA. Classical test theory, however, can only incorporate a single general error term. G-Theory, by contrast, was designed to differentiate between multiple sources of error (Cronbach et al., 1972; Shavelson & Webb, 1991; Brennan, 2001). That is, the error component in G-Theory is subdivided into a systematic component, which consists of error attributable to known sources of variation (so-called *facets*, such as *participants* and *items*) and an un-systematic or ‘residual’ component, which consists of error attributable to random and/or unidentified effects as well as the interaction of known sources of variation (which cannot be disentangled, see Shavelson & Webb, 1991). Thus, G-Theory acknowledges that inter-item and inter-participant variation are potential sources of error in the generalization of acceptability measurement and, furthermore, it takes into account that there will always be residual error blurring the picture.

Mathematically, reliability in G-Theory is defined as it is in classical test theory, i.e., as the relative contribution of the true score component to the total variance. However, G-Theory is particularly suitable for reliability analysis in the field of experimental syntax (see Schoenmakers & van Hout, 2024), not only because it can accommodate multiple facets, but also because it can handle the nested experimental designs common to the field (i.e., the Latin Square design, according to which not all participants see all items in all conditions, with the stimulus sentences nested in different conditions). The metric for reliability used in G-Theory is the *generalizability coefficient*, or ‘G-coefficient’, represented by the Greek letter ρ^2 . The G-coefficient permits relative inferences about rank-order, like the reliability of our datasets. Reliability is poor when $\rho^2 < 0.5$, moderate when $0.5 < \rho^2 < 0.75$, good when $0.75 < \rho^2 < 0.9$, and excellent when $\rho^2 \geq .9$ (Koo & Li, 2016).⁵

3. A case study: island effects

The present paper takes experiments that investigate island effects as a case study. Island effects are commonly defined as the super-additivity that arises when a long-distance dependency originates in a particular, so-called ‘island’ construction (Sprouse & Villata, 2021). To test for island effects, experimental researchers typically use a factorial design in which the factors *distance* (short- vs. long-distance movement) and *construction* ($\pm$ island) are crossed, a design that was originally proposed in Sprouse (2007); a sample item which follows this design is given in (1). The rationale behind this particular design is that both the formation of a long-distance dependency and the presence of a complex structure may independently affect acceptability in a negative way, but the *island effect* refers to the decrease in acceptability that remains when the variation independently attributable to the two factors is accounted for. Island effects are thus represented by an interaction effect. The effect size of these interactions is commonly reported as the *differences-in-differences score* (Maxwell & Delaney, 2004), or simply DD-score, and can be calculated by subtracting the difference between (1a–c) from the difference between (1b–d), which isolates the additional effect not accounted for by the two fixed factors (Sprouse & Villata, 2021).

- | | |
|---|--------------------|
| (1) a. Who ____ thinks that the lawyer forgot his briefcase at the office? | [short, no island] |
| b. What do you think that the lawyer forgot ____ at the office? | [long, no island] |
| c. Who ____ worries if the lawyer forgot his briefcase at the office? | [short, island] |
| d. What do you worry if the lawyer forgot at the office? | [long, island] |
- (Sprouse et al., 2016: 318)

The calculation of DD-scores representing island effect sizes is thus based on each of the conditions in the paradigm. Given that linguistic data are by definition noisy (i.e., associated with unknown and

⁵ The guidelines in Koo and Li (2016) are actually for the ICC, but they can also be applied to the G-coefficient, which is essentially a “stepped-up intraclass correlation coefficient” (Brennan, 2001: 35).

unidentified variation), reliable measurement is essential for a proper interpretation of the DD-score. Yet, reliability coefficients are rarely, if ever, reported in experimental syntactic work, even though researchers do tend to make explicit comparisons of island effect sizes across experiments, configurations, dependency types, and languages (see e.g., the discussion of Figure 1 in Pañeda et al., 2020 or Section 9.4 in Sprouse & Villata 2021). As alluded to in the introduction section, a large enough number of items should be tested for reliable measurement. Most modern experiments test more than a single item, each of which is represented by four minimally different variants, as in (1) (see also Cowart 1997: §3.3 on the design of experimental materials). The items are then usually distributed according to a Latin Square design: each participant is presented with only one variant of each particular item. Thus, a first participant sees the first sentence in condition A, a second participant sees this same sentence in condition B, a third in condition C, a fourth in condition D, et cetera. The first participant is then presented with *the second sentence* in condition B, *the third sentence* in condition C, and *the fourth sentence* in condition D. The Latin Square design is illustrated in Table 1. Following this, experiments can be constructed such that every participant sees an equal number of items in a particular condition. With 24 items, for example, each participant would be presented with six items in each condition.

Table 1. Latin Square design, with the particular conditions of a factorial 2×2 design (A, B, C, D) represented in the cells, and with ellipses indicating the recurring pattern for additional items and participants.

	Item 1	Item 2	Item 3	Item 4	...
Participant 1	A	B	C	D	...
Participant 2	B	C	D	A	...
Participant 3	C	D	A	B	...
Participant 4	D	A	B	C	...
...	...	...	...	...	...

But the question remains how many items should be used for reliable experimentation. As it turns out, there is no general consensus on this. Sprouse and Schütze (2014: 39) submit that “it is desirable to create multiple lexicalizations of each condition (ideally eight or more)”. In his discussion on the construction of experimental syntactic questionnaires, Cowart (1997: 92) provides an example of an experiment that follows a 2×2 design with six instances of each sentence type (i.e., 24 items in total). Goodall (2021: 13), however, refers to an item set with four sentences per condition in a 2x2 design (i.e., sixteen sentences in total) as “a full set [...] with only a minimal possibility of error,” and Jon Sprouse’s (2018) course slides maintain that two items per condition suffices:

“For basic acceptability judgment experiments I generally recommend that you present 2 items per condition per participant. So for a 2x2 design, that means you need 8 items per condition. I think that 8 items per condition is also sufficient to make (non-statistical) claims about the generalizability of the result to multiple items. So this is a nice starting point for most designs. Of course, if you have reason to believe that participants will make errors with the items, you should present more than 2 items per condition. Similarly, if you need to demonstrate that the result generalizes to more than 8 items, by all means, use more than 8 items. These are just good starting points for basic acceptability judgment experiments.”

(Sprouse, 2018: slide 62)

Experiments investigating island effects consequently vary in terms of the number of items tested, as will become clear in the next two subsections. These subsections more specifically calculate the reliability of a collection of existing datasets (based on generalizability studies) and determine what number of items would yield ‘a minimal possibility of error’ and permit ‘claims about the generalizability of the result to multiple items’ (based on decision studies).

Before moving on to these computations, though, notice that the sample item in (1) displays the factorial design with a (bare) wh-dependency and an adjunct island: the fillers are bare wh-phrases (*who, what*) and the extraction domain in the island conditions are adjunct clauses (here, *if the lawyer forgot (his briefcase) at the office*). But island effects have been tested with other types of dependencies and configurations as well. Other dependency types that have been tested in experimental work include relativization, topicalization, and wh-extraction with complex fillers (e.g., *which briefcase*). Other island configurations that have been experimentally tested include complex NPs, embedded questions, and relative clauses. The resulting combinations of these variables have been tested in a range of languages using particularly similar designs and operationalizations: researchers have conducted translated adaptations of the experimental design in (1), keeping the instrument of measurement relatively constant. Datasets on island effects are therefore particularly suitable as a case study for the reliability analyses performed in this paper.

Note finally that, while various papers have been published that report significant island effects, it is quite possible that many other island experiments have been conducted but were never published because they yielded non-significant or uninterpretable results. Further, as noted, the data from those experiments that *were* published might not be entirely reliable because of inexplicable variation among participants and/or items. This is quite an issue, as the island effect sizes may then be overestimated and, as Loken and Gelman (2017: 584) put it, “statistical significance conveys very little information when measurements are noisy.” The next subsection therefore performs reliability analyses on selected datasets from island experiments to further explore their variation patterns.

3.1. Data selection

A total of 55 distinct datasets was initially included in the analysis, taken from Almeida (2014), Sprouse et al. (2016), Kush et al. (2018, 2019), Pañeda and Kush (2021), Kobzeva et al. (2022), Rodríguez and Goodall (2023), and Schoenmakers and Stoica (2024). However, the default formula used in the models (Section 3.2) yielded singularity issues in three datasets (the English set from Almeida, 2014, the adjunct island set with complex fillers from Kush et al., 2018, and the complex NP dataset from Rodríguez & Goodall, 2023). Singularity issues occur when there is extreme collinearity in the data, so that a given regression coefficient cannot be accurately estimated. That is to say, collinearity is an effect of the imprecision of measurement and it thus interacts with sample size as well (see O’Brien, 2007). As the variance associated with the item facet could not be estimated in these datasets, they were removed from further analysis. The reliability analyses were performed on the remaining 52 datasets.

The selected studies all used the factorial design in (1) to investigate island sensitivity in six different languages, using seven-point scales ranging from (very) bad to (very) good to collect acceptability scores (the details of each experimental design can be found in Table 2). Each study tested a slightly different selection of island types, but four types were commonly tested across experiments and were therefore included in the present analyses: (conditional) adjunct islands, complex NP islands, embedded *whether* islands, and/or relative clause (RC) islands (note that while the operationalization of the experiments in this sample was clearly not identical across the board, the differences are not an issue for the purposes of the present paper, because no direct comparisons are made). Furthermore, different dependency types were tested in the selected studies, *viz.*, wh-movement of bare wh-phrases and/or full wh-DPs, topicalization, and/or relativization. Note, finally, that the datasets involved in the present analyses were reduced to the factorial design described in Sprouse (2007). Thus, while e.g., Kobzeva et al. (2022) tested both wh- and RC-dependencies in the same experiment, here these are represented as two distinct datasets: one with wh-dependencies and one with RC-dependencies.

The selected studies differed in the numbers of items tested per dataset. Rodríguez and Goodall (2023) tested the largest number in their experiments, with four items per condition. Next in line are Pañeda and Kush (2021), who tested three items per condition in the *whether* island item sets, but two items per condition for the adjunct and RC island item sets (because these served as controls for their

purposes). Sprouse et al. (2016), Kush et al. (2018, 2019), Kobzeva et al. (2022), and Schoenmakers and Stoica (2024) tested two items per condition, and, finally, Almeida (2014) tested one item per condition.

RC islands were tested in Norwegian and Spanish, but the type of clauses differed between experiments. Kush et al. (2018, 2019), Pañeda and Kush (2021), and Rodríguez and Goodall (2023) tested RC islands where the head was an argument of the matrix verb, as in (2). Kobzeva et al. (2022), by contrast, tested existential relative clauses, as in (3), because corpus data show that such RCs are more common. The main difference is that the sentences in (3) are typically used to introduce a new discourse referent and provide additional (new) information in the RC.

- (2) a. { **Hva** / **Hvilken film** } snakket regissøren med [et par kritikere som hadde stemt
what / which film spoke director.DEF with a few critics who had voted
på ____] ?
for
'What/Which film did the director speak with [a few critics that had voted for ____]?'
(Kush et al., 2018)
- b. **Antidepressiva** kjenner de til [mange psykologer som ville anbefale ____].
antidepressants know they to many psychologists who would recommend
'Antidepressants they know [many psychologists who would recommend ____]?'
(Kush et al., 2019)
- c. ¿**A quién** conociste [al ____ sindicato que apoya ____] ?
to who knew.2SG to.the union that supports
'(Of) who(m) did you know [the union that supports ____] ?'
(Rodríguez & Goodall, 2023)
- d. ¿**Qué proyecto** tenía el jefe [unos empleados que habían presentado ____] ?
what project had the chief some employees that have presented
'Which project did the chief have [some employees that had presented ____] ?'
(Pañeda & Kush, 2021)
- (3) a. **Hvilket øl** var det [mange som bestilte ____] ?
which beer was it many who ordered
'Which beer were there [many people who ordered ____] ?'
- b. Det var **ølet** som det var [mange som bestilte ____].
that was beer.DEF that it was many who ordered
'That was the beer that there were [many people who ordered ____].' (Kobzeva et al., 2022)

Some of the selected studies also tested additional island types, which were not included in the present analyses. Pañeda and Kush (2021) tested *where* and *when* interrogative islands, Kobzeva et al. (2022) tested *what* and *where* interrogative islands, and Rodríguez and Goodall (2023) tested *when* and *where* interrogative islands. These 'alternative' embedded question islands show different effects from their more commonly tested counterparts headed by *whether* and were therefore excluded from analysis. Further, some studies also tested for subject island effects. Subject island effects, however, have been tested using vastly different designs (see e.g., Kush et al., 2018 for discussion) and were consequently excluded from the analyses as well. Regarding the tested dependency types, Almeida (2014) and Rodríguez and Goodall (2023) also tested cases of left dislocation in Brazilian Portuguese and Spanish. Such configurations are normally not taken to result from movement, however, so this dependency type is not expected to yield island effects. The clitic left dislocation cases were therefore excluded from the analyses as well.

3.2. Data analysis

R (R Core Team, 2024, version 4.4.0) and the *gtheory* package (Moore, 2016) were used to run generalizability and decision studies on the selected datasets. All G-studies and D-studies were based on linear mixed effects models which included the fixed effects of *distance* and *construction* and their interaction, as well as random by-participant and by-item intercepts (following Huebner & Lucht, 2019). The last two columns in Table 2 display the reliability (ρ^2) of each dataset as well as the reliability if the number of items were increased to 24, as determined in a series of D-studies. Good reliability, with ρ^2 -values above 0.75 (Koo & Li, 2016), is reached in fifteen of the 52 (28.8%) original datasets. The predictive ability of these studies is generally moderate with an average G-coefficient of 0.66. If the number of items in each experiment were increased to 24, however, the number of datasets with good reliability would increase to 46 of 52 (88.5%), with an average G-coefficient of 0.84. The increase in reliability per added item is visually presented for each dataset in Figures 1–12. These figures illustrate particularly clearly that a relatively minor adjustment to the experimental design can greatly improve the reliability of the dataset. This effect is especially pronounced when the number of items is small, as even a slight increase can enhance the reliability considerably.

Nonetheless, the D-studies predict that with an increase to 24 items, six datasets would still fall short of the criterion for good reliability. Among them, Kobzeva et al.'s (2022) RC island dataset with (complex) wh-movement and Rodríguez and Goodall's (2023) adjunct island dataset with (bare) wh-movement would come close to the benchmark for good reliability, with G-coefficients of 0.73 and 0.72, respectively. Sprouse et al.'s (2016) Italian adjunct island dataset with an RC-dependency can be considered an outlier, however, with a G-coefficient of 0.59. Notably, this is the only dataset taken from Sprouse et al. (2016) that did not produce a significant island effect. This non-significance could be indicative of a Type II error, as a low number of items lowers power estimates just like a low number of participants does (see Sprouse & Almeida, 2017).⁶ The results from the D-study performed on this dataset indicate that 67.5% of the variation is part of the residual error component. For comparison, the proportion of residual variation in the (Italian) data with adjunct islands and a wh-dependency was 28.7%; the proportion of residual variation in the English counterpart of the dataset (i.e., with an RC-dependency) was 23.8%.⁷ The variation in the Italian version of this dataset was thus for the most part not uniquely attributable to participants or items, and so an increase of the number of items would not improve reliability much.

Finally, three of Kush et al.'s (2019) datasets from Experiment 2 would not achieve good reliability with an increase to 24 items, specifically, those datasets with a topicalization dependency and adjunct, complex NP, or *whether* islands (with G-coefficients of 0.69, 0.62, and 0.69, respectively). Participants in this Experiment 2 were asked to read a brief preamble before assessing the acceptability of the target sentence. Kush et al. also conducted the same experiment without contextualization (Experiment 1), which resulted in much more reliable datasets (see Table 2). However, the authors still noted considerable intertrial variability and inconsistency in these judgments. They suggest, with regard to the embedded question islands in particular, that participants may adapt or recalibrate after seeing a first island violation, so that they are more amenable to it upon a second encounter. The authors conclude that a larger number of observations per participant are needed to investigate this further – a claim that is corroborated by the D-studies reported on here, see Figure 8. Still, the discrepancy between their two experiments is rather striking. It is conceivable that participants in Experiment 2 were not in fact helped by the discourse context, but were instead distracted or confused by it, and consequently performed in a less consistent manner (at least in the cases with an island violation, see Kush et al., 2019). However,

^{6.} Both statistical power and reliability are defined in relation to the total variance of the dataset. However, statistical power is computed irrespective of the partitioning of the observed scores into a true and error component whereas reliability is a function of the ratio between the true component and the observed component. The reliability coefficient can consequently vary while statistical power remains unchanged, depending on the relative contributions of these components (see Zimmerman & Zumbo, 2015).

^{7.} This information as well as the variance components associated with the individual facets can be found in the analysis output in the OSF repository.

the target sentences in Pañeda and Kush's (2021) experiments, which tested Spanish wh-dependencies in slightly different configurations, were preceded by a context sentence as well, and yet these datasets are associated with much better reliability. The impact of contextualization thus seems to remain a methodological question that warrants further investigation. To explore this, however, it is crucial to collect larger numbers of observations per participant (cf. Kush et al., 2018, 2019).

4. Conclusion

Ensuring reliability in the social sciences is crucial, because human measurement is inherently imprecise. In the study of island effects, variation is already widespread at the surface, arising from cross-linguistic and cross-configurational differences, as well as from differences in experimental design and stimuli. But to make meaningful comparisons of experimental data and island effect sizes in particular, an important first step is to calculate the reliability of the datasets.

This paper explored the variation patterns in 52 distinct datasets from relatively similar experimental studies (i.e., translated adaptations) on island sensitivity, using Generalizability Theory to estimate the variance components of each dataset. The findings reveal that the most datasets exhibit only moderate reliability. This raises a critical concern: when island effect sizes are reported without considering reliability, and especially when DD-scores are compared, there is a risk of overestimating the effect sizes due to measurement imprecision (i.e., when reliability is low). The present paper argued that even a modest increase in the number of experimental items can substantially improve the reliability of datasets from island experiments. A series of D-studies, which are designed to estimate a measure in which error is minimized, show that measurement can be made sufficiently precise when 24 items are tested in a factorial 2×2 design (i.e., six items per condition), distributed under a Latin Square. This is a practical and achievable adjustment to island experimental design: instead of conducting small-scale acceptability judgment experiments on multiple island types, researchers could focus on testing single island types in larger experiments, thereby enhancing the precision of measurement and ultimately strengthening the validity of their findings.

Table 2. Selected dataset specifics, with dependency type, island type, number of participants n , number of items k , generalizability coefficients, and calculated generalizability coefficients for $k=24$. Note that the number of items is determined based on the original 2×2 design for the study type (e.g., for RC dependencies only). The target sentences in Kush et al. (2019, Experiment 2) and Pañeda and Kush (2021) were preceded by a context sentence.

Language	Source	Dep. type	Island type	n	k	ρ^2	$\rho^2 (k=24)$
Br. Portuguese	Almeida (2014)	bare <i>wh</i>	Interrogative	38	4	0.53	0.87
		topicalization	Interrogative	38	4	0.45	0.83
English	Sprouse et al. (2016)	bare <i>wh</i>	Adjuncts	55	8	0.73	0.89
			Complex NP	55	8	0.57	0.80
			Interrogative	55	8	0.78	0.91
		complex <i>wh</i>	Adjuncts	56	8	0.65	0.85
			Complex NP	54	8	0.71	0.88
			Interrogative	56	8	0.78	0.91
		RC	Adjuncts	55	8	0.76	0.90
			Complex NP	55	8	0.81	0.93
			Interrogative	55	8	0.82	0.93
Norwegian	Kush et al. (2018)	bare <i>wh</i> (exp.1)	Adjuncts	94	8	0.53	0.77
			Complex NP	94	8	0.70	0.87
			Interrogative	94	8	0.73	0.89
		bare <i>wh</i> (exp.2)	Adjuncts	46	8	0.63	0.84
			Complex NP	46	8	0.51	0.76
			Interrogative	46	8	0.80	0.92
		complex <i>wh</i>	RC (argument of matrix verb)	46	8	0.81	0.93
			Complex NP	66	8	0.61	0.82
			Interrogative	66	8	0.78	0.91
			RC (argument of matrix verb)	66	8	0.69	0.87

Language	Source	Dep. type	Island type	n	k	ρ^2	ρ^2 (k=24)
Norwegian (cont.)	Kush et al. (2019)	topicalization (exp. 1)	Adjuncts	32	8	0.68	0.87
			Complex NP	32	8	0.59	0.81
			Interrogative	32	8	0.65	0.85
			RC (argument of matrix verb)	32	8	0.72	0.89
		topicalization (exp. 2)	Adjuncts	31	8	0.43	0.69
			Complex NP	31	8	0.36	0.62
			Interrogative	31	8	0.43	0.69
			RC (argument of matrix verb)	31	8	0.72	0.89
	Kobzeva et al. (2022)	complex <i>wh</i>	Adjuncts	93	8	0.70	0.87
			RC (existential constructions)	93	8	0.48	0.73
		RC	Adjuncts	93	8	0.66	0.85
			RC (existential constructions)	93	8	0.64	0.84
Italian	Sprouse et al. (2016)	bare <i>wh</i>	Adjuncts	47	8	0.71	0.88
			Complex NP	47	8	0.68	0.86
			Interrogative	50	8	0.76	0.90
		RC	Adjuncts	47	8	0.32	0.59
			Complex NP	47	8	0.72	0.89
			Interrogative	49	8	0.65	0.85
Romanian	Schoenmakers and Stoica (2024)	bare <i>wh</i>	Adjuncts	103	8	0.56	0.79
			Complex NP	103	8	0.63	0.84
			Interrogative	101	8	0.78	0.91

Language	Source	Dep. type	Island type	n	k	ρ^2	ρ^2 (k=24)
Spanish	Pañeda and Kush (2021)	complex <i>wh</i>	Adjuncts (Exp. 1)	47	8	0.75	0.90
			Adjuncts (Exp. 2)	49	8	0.63	0.84
			Interrogative (whether, <i>know</i>)	47	12	0.82	0.90
			Interrogative (whether, <i>ask</i>)	47	12	0.75	0.86
			RC (Exp. 1)	47	8	0.61	0.83
	Rodríguez and Goodall (2023)	bare <i>wh</i>	RC (Exp. 2)	49	8	0.51	0.75
			Adjuncts	48	16	0.63	0.72
			Interrogative (Exp. 1)	48	16	0.82	0.87
			Interrogative (Exp. 2)	48	16	0.71	0.78
			RC (argument of matrix verb)	48	16	0.73	0.80

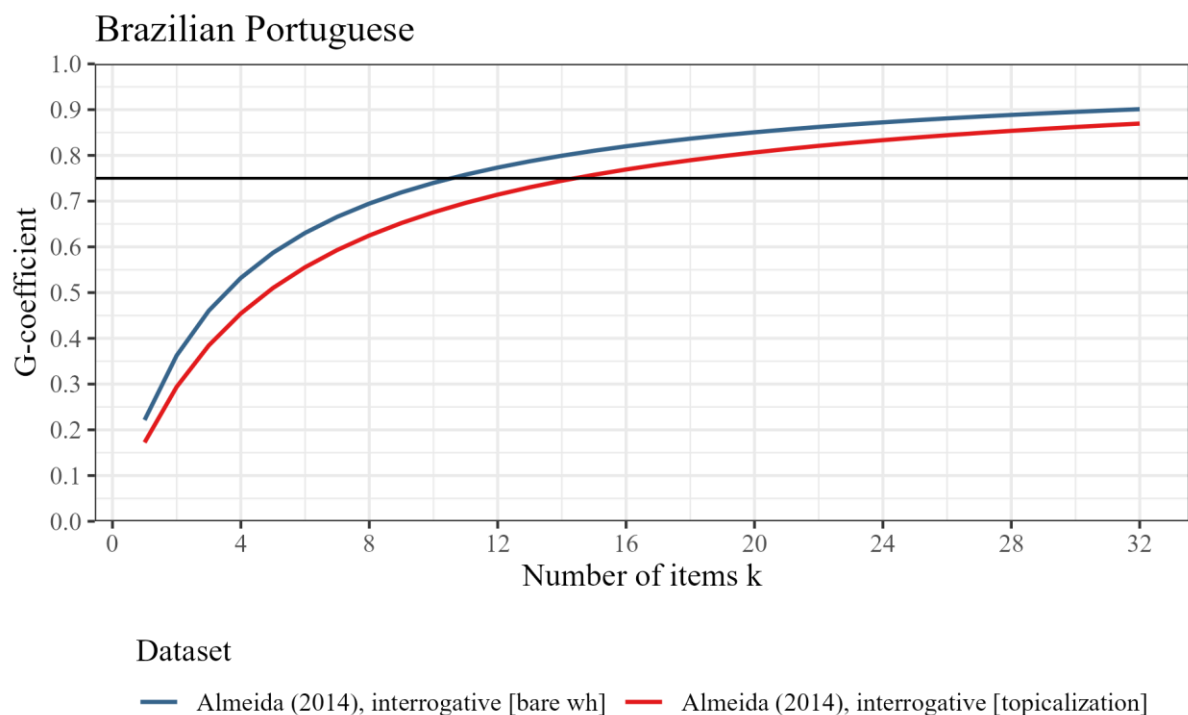


Figure 1. Generalizability coefficient (reliability) by number of items tested, based on two datasets from Almeida (2014), for Brazilian Portuguese. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

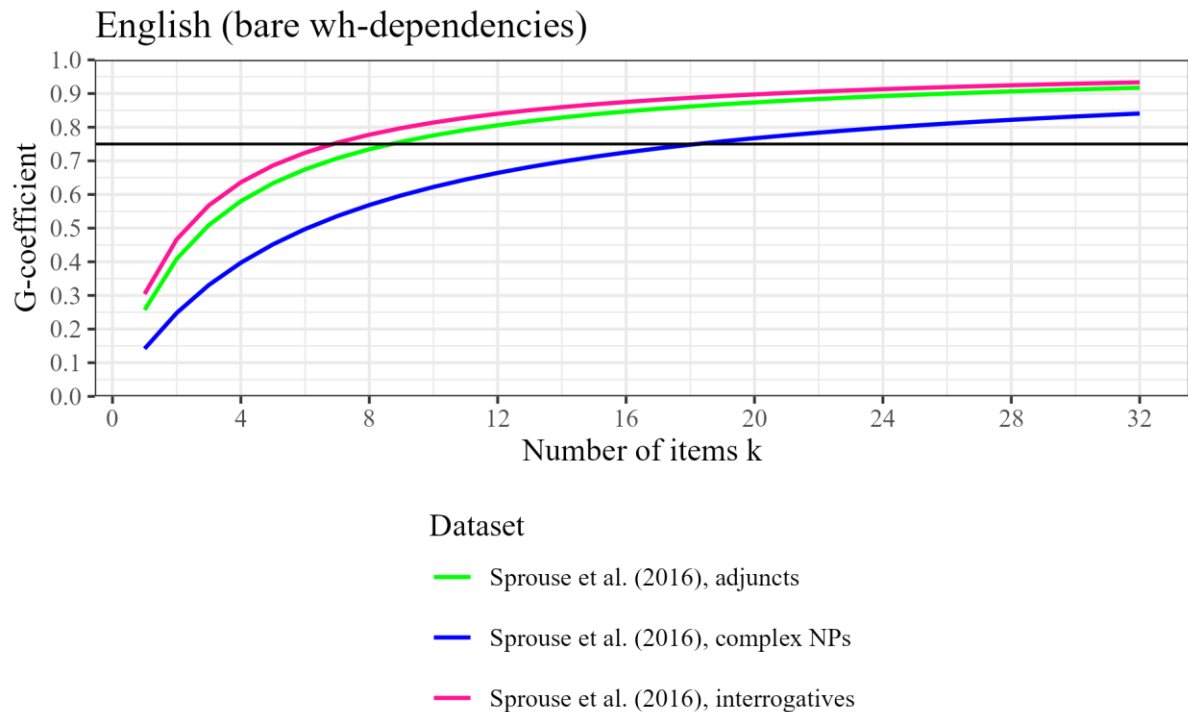


Figure 2. Generalizability coefficient (reliability) by number of items tested, based on three datasets from Sprouse et al. (2016) with bare *wh*-dependencies, for English. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

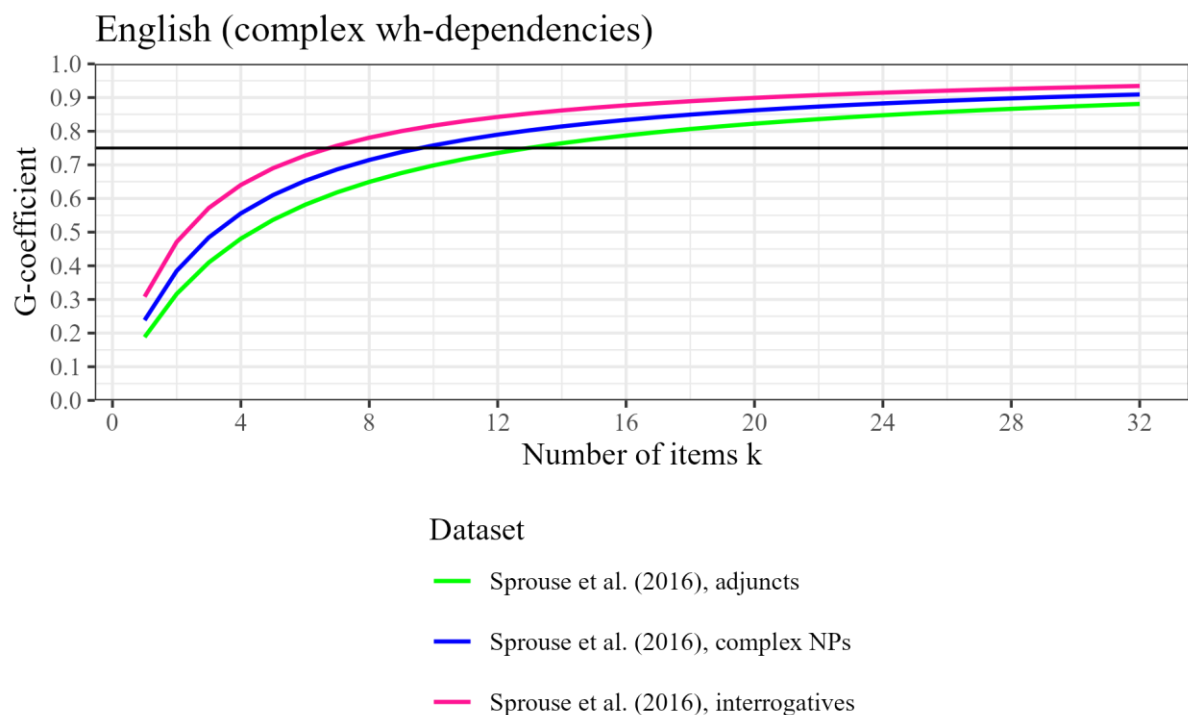


Figure 3. Generalizability coefficient (reliability) by number of items tested, based on three datasets from Sprouse et al. (2016) with complex *wh*-dependencies, for English. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

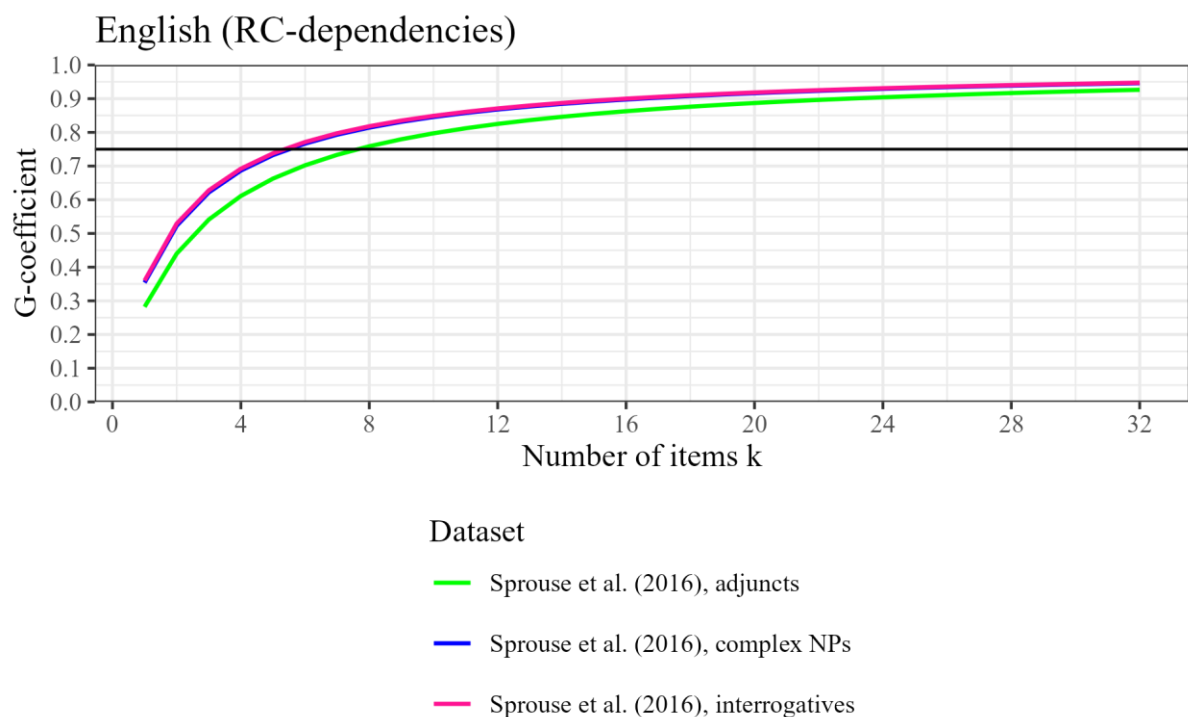


Figure 4. Generalizability coefficient (reliability) by number of items tested, based on three datasets from Sprouse et al. (2016) with RC-dependencies, for English. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

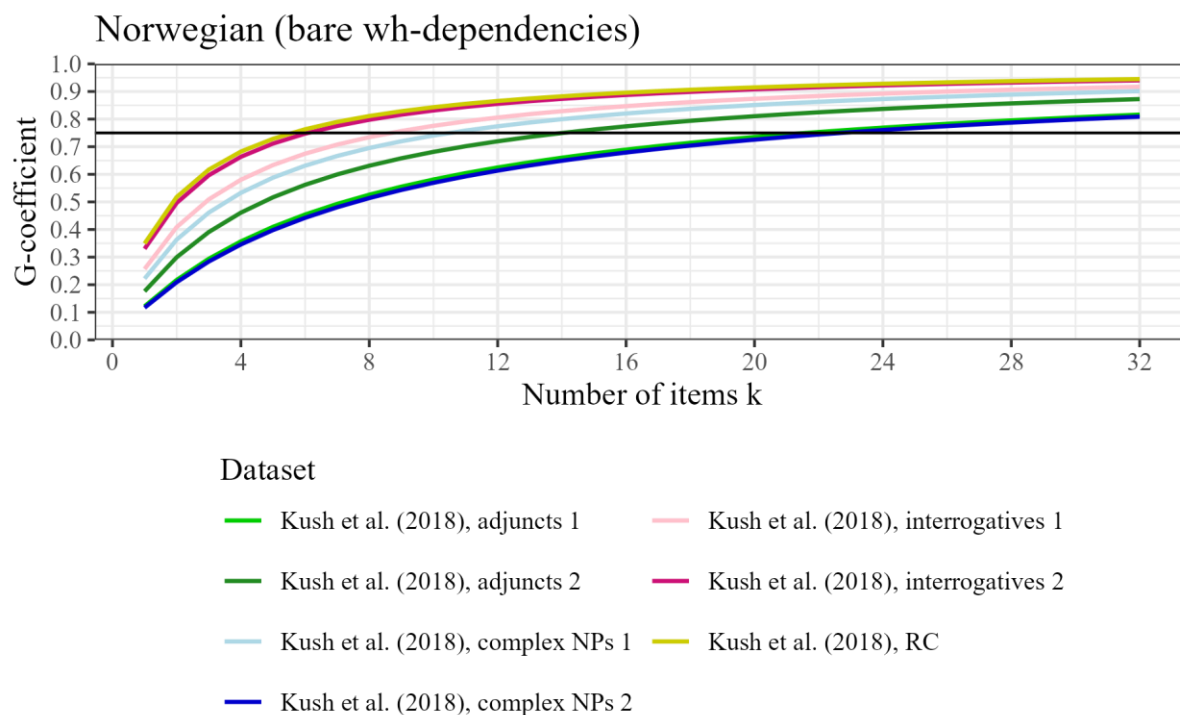


Figure 5. Generalizability coefficient (reliability) by number of items tested, based on seven datasets from Kush et al. (2018) with bare *wh*-dependencies, for Norwegian. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

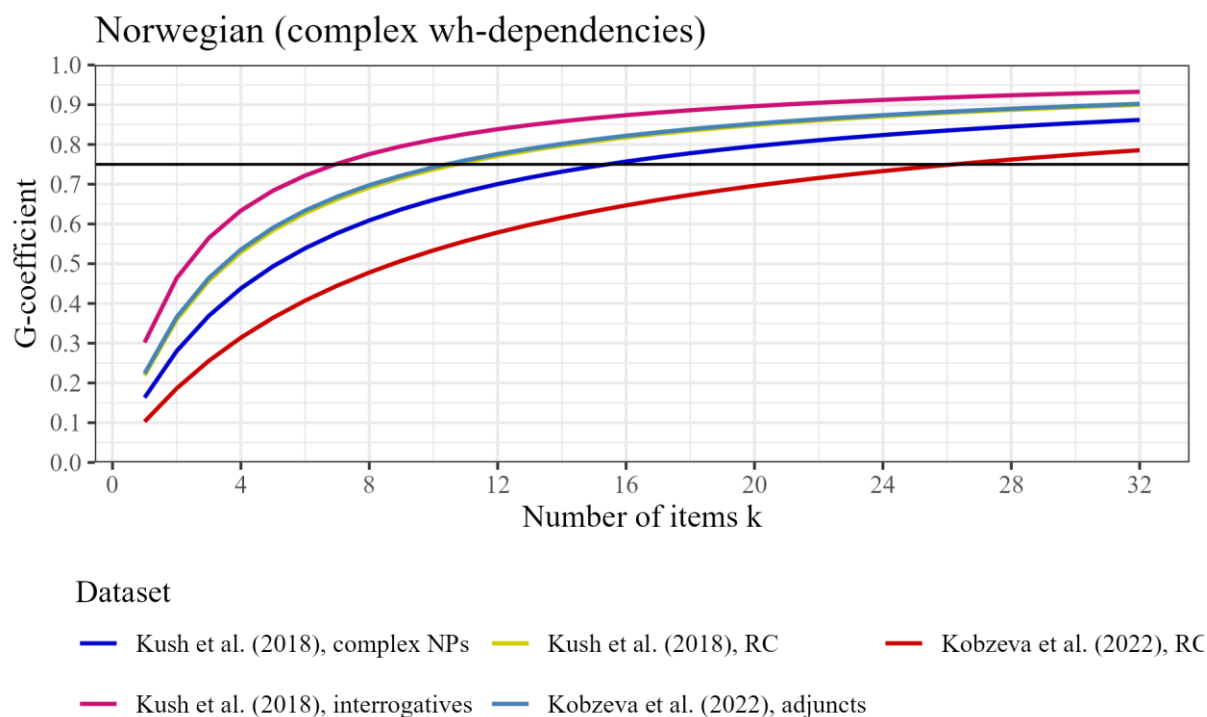


Figure 6. Generalizability coefficient (reliability) by number of items tested, based on five datasets from Kush et al. (2018) with complex *wh*-dependencies, for Norwegian. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

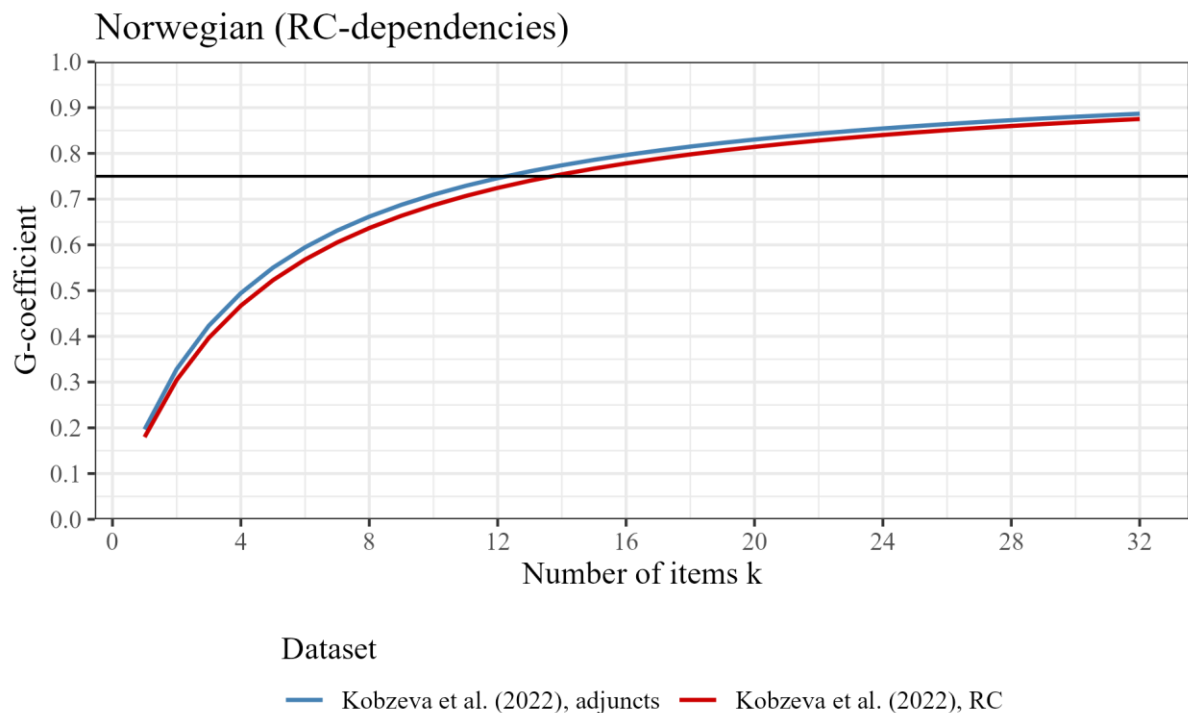


Figure 7. Generalizability coefficient (reliability) by number of items tested, based on two datasets from Kobzeva et al. (2022) with RC-dependencies, for Norwegian. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

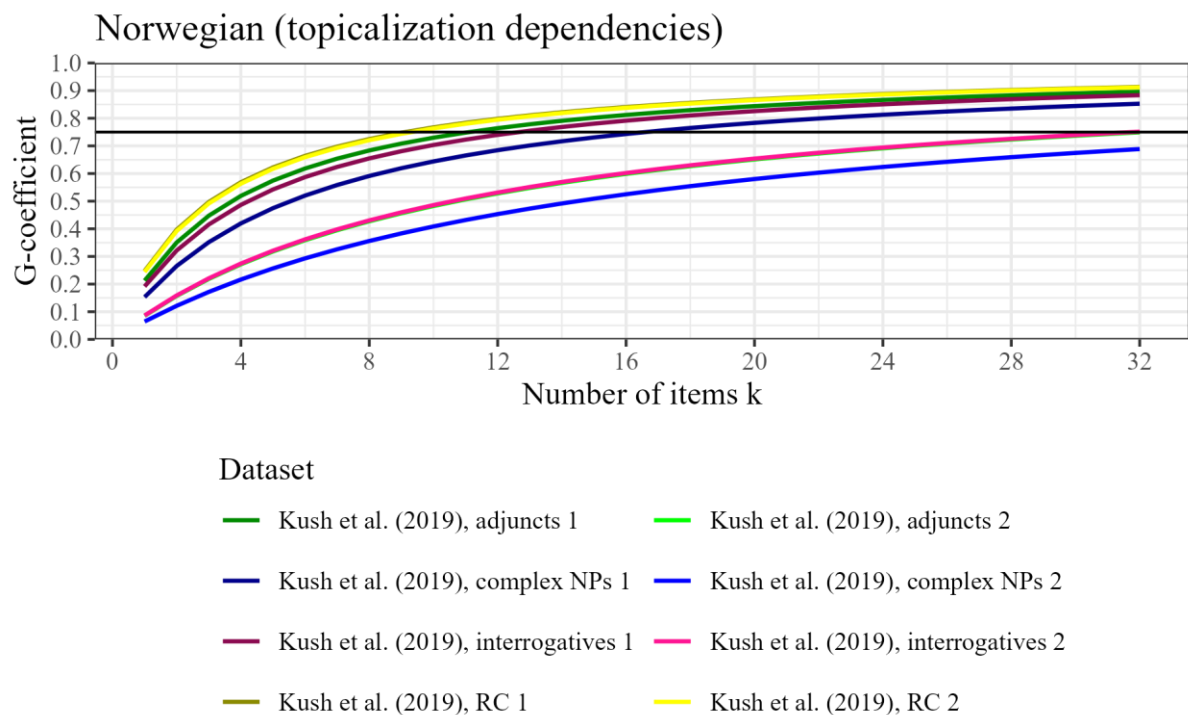


Figure 8. Generalizability coefficient (reliability) by number of items tested, based on eight datasets from Kush et al. (2019) with topicalization dependencies, for Norwegian. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

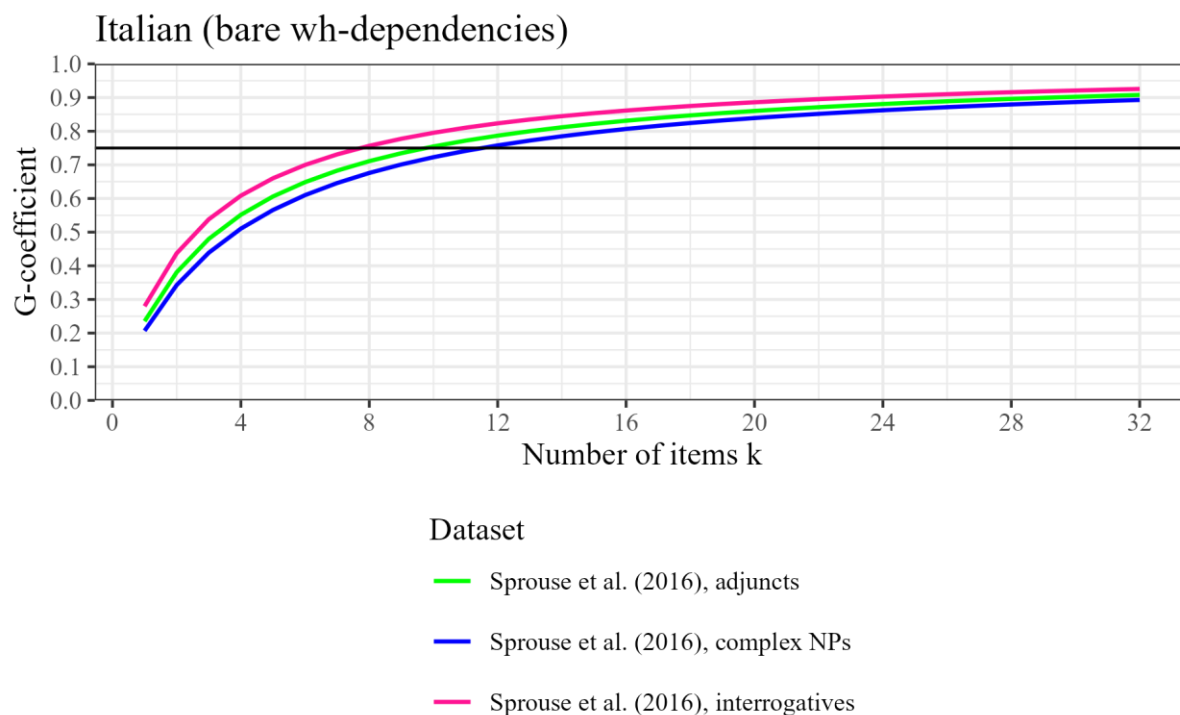


Figure 9. Generalizability coefficient (reliability) by number of items tested, based on three datasets from Sprouse et al. (2016) with bare *wh*-dependencies, for Italian. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

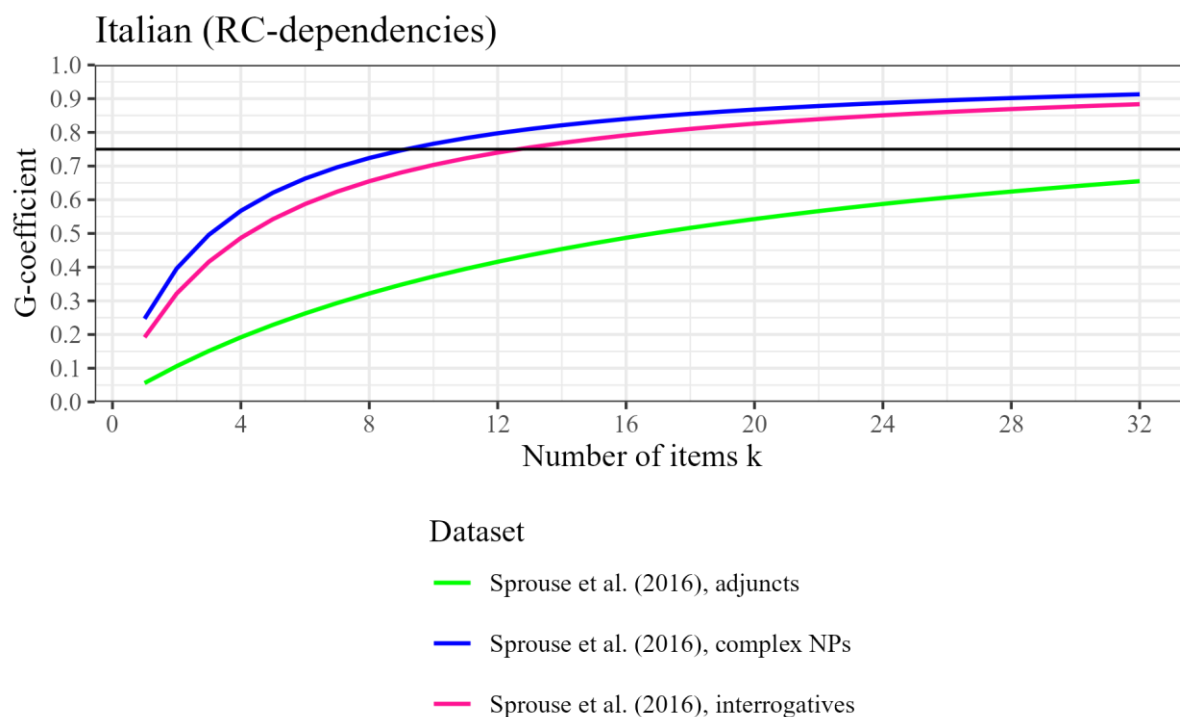


Figure 10. Generalizability coefficient (reliability) by number of items tested, based on three datasets from Sprouse et al. (2016) with RC-dependencies, for Italian. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

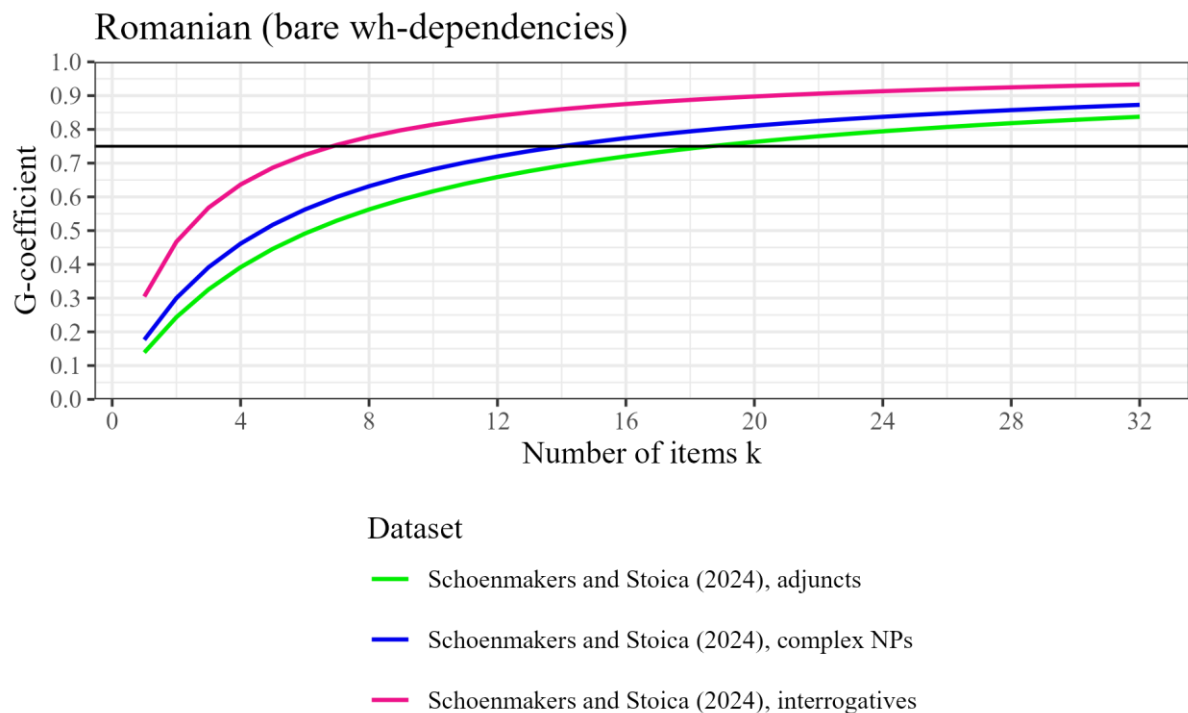


Figure 11. Generalizability coefficient (reliability) by number of items tested, based on three datasets from Schoenmakers and Stoica (2024) with bare *wh*-dependencies, for Romanian. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

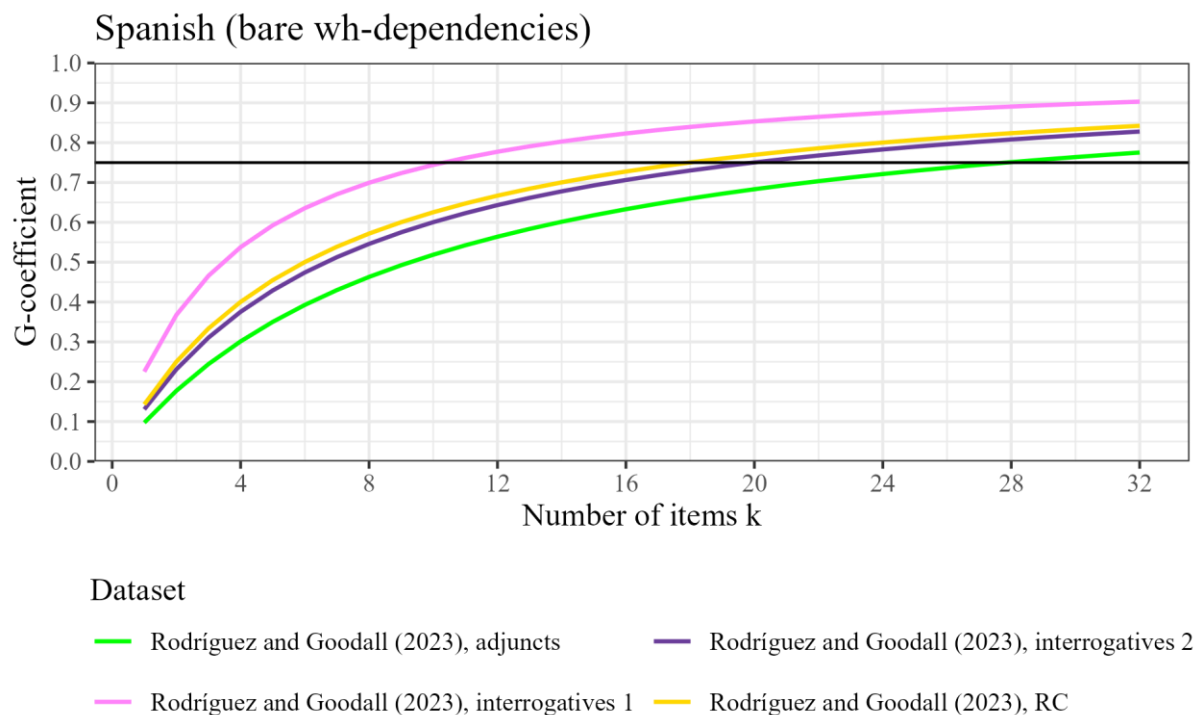


Figure 12. Generalizability coefficient (reliability) by number of items tested, based on four datasets from Rodríguez and Goodall (2023) with bare *wh*-dependencies, for Spanish. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

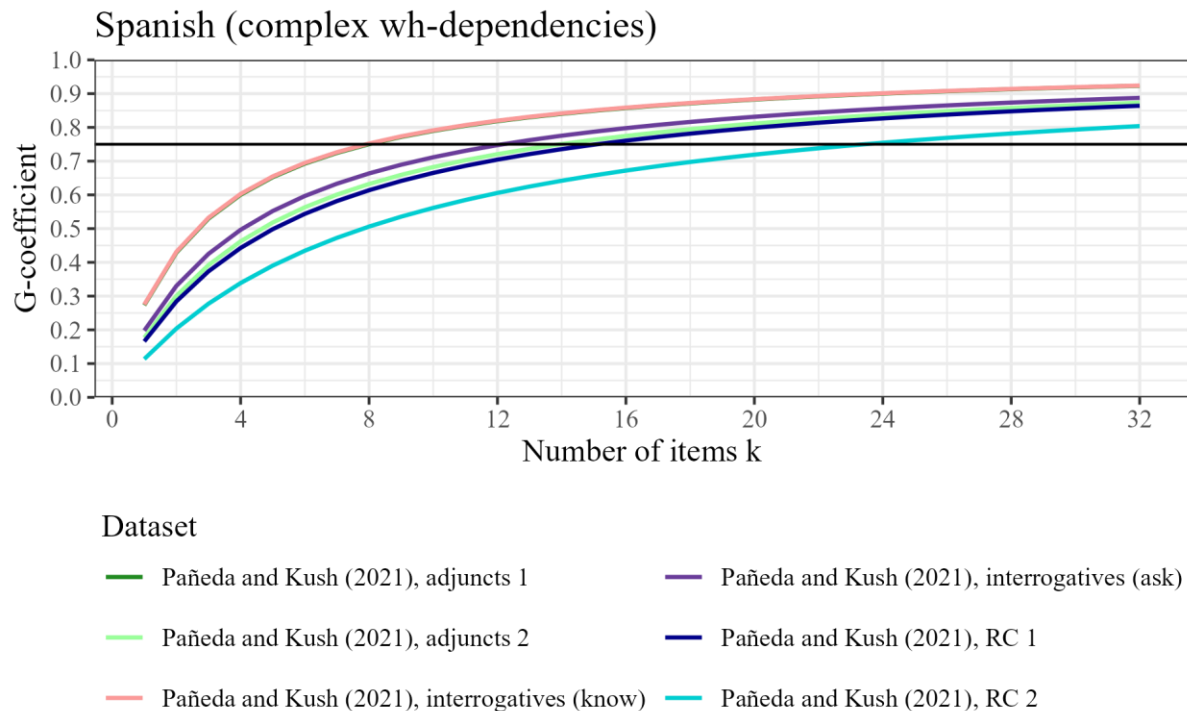


Figure 12. Generalizability coefficient (reliability) by number of items tested, based on six datasets from Pañeda and Kush (2021) with complex *wh*-dependencies, for Spanish. The horizontal line represents the threshold for good reliability at $\rho^2 = 0.75$.

Data Availability Statement

New data have not been collected for this paper; all datasets that were analyzed have been collected from existing online repositories or were personally sent to the author. Stripped versions of all the datasets (ratings, participant and item numbers, and conditions) as well as the analysis code output can be found in a dedicated OSF repository (anonymized view-only link: https://osf.io/k4ndm/?view_only=325003b150794b678eeca050f6336a3). Permission to share data from existing datasets that were not previously shared online was granted by the first authors of the original papers.

References

- Almeida, D. (2014). Subliminal wh-islands in Brazilian Portuguese and the consequences for syntactic theory. *Revista de ABRALIN*, 13(2), 55–93. <https://doi.org/10.5380/rabl.v13i2.39611>
- Brennan, R. (2001). *Generalizability Theory*. Springer. https://doi.org/10.1007/978-1-4757-3456-0_6
- Brysbaert, M. (2019). How many participants do we have to include in properly powered experiments? A tutorial of power analysis with reference tables. *Journal of Cognition*, 2(1), Article 16. <https://doi.org/10.5334/joc.72>
- Christianson, K., & Dempsey, J. & Tsiola, A. & Goldshtein, M. (2022). What if they're just not that into you (or your experiment)? On motivation and psycholinguistics. *Psychology of Learning and Motivation*, 76, 51–88. <https://doi.org/10.1016/bs.plm.2022.03.002>
- Cowart, W. (1997). *Experimental syntax: Applying objective methods to sentence judgments*. SAGE.
- Cronbach, L., Gleser, G., Nanda, H., Rajaratnam, N. (1972). *The dependability of behavioral measurements: Theory of generalizability for scores and profiles*. John Wiley.

- Dahl, Ö. (1979). Is linguistics empirical? A critique of Esa Itkonen's Linguistics and metascience. In T. Perry (Ed.), *Evidence and argumentation in linguistics* (pp. 13–45). De Gruyter.
<https://doi.org/10.1515/9783110848854-008>
- Dempsey, J., & Christianson, K. (2022). Referencing context in sentence processing: A failure to replicate the strong interactive mental models hypothesis. *Journal of Memory and Language*, 125, Article 104335.
<https://doi.org/10.1016/j.jml.2022.104335>
- Gelman, A. (2018). Ethics in statistical practice and communication: Five recommendations. *Significance*, 15(5), 40–43. <https://doi.org/10.1111/j.1740-9713.2018.01193.x>
- Gibson, E., & Fedorenko, E. (2013). The need for quantitative methods in syntax and semantics research. *Language and Cognitive Processes*, 28(1–2), 88–124. <https://doi.org/10.1080/01690965.2010.515080>
- Goodall, G. (2021). Sentence acceptability experiments: What, how, and why. In G. Goodall (Ed.), *The Cambridge handbook of experimental syntax* (pp. 5–39). Cambridge University Press.
<https://doi.org/10.1017/9781108569620.002>
- Grieve, J. (2021). Observation, experimentation, and replication in linguistics. *Linguistics*, 59(5), 1343–1356.
<https://doi.org/10.1515/ling-2021-0094>
- Huebner, A., & Lucht, M. (2019). Generalizability Theory in R. *Practical Assessment, Research & Evaluation*, 24(5), Article 5. <https://doi.org/10.7275/5065-gc10>
- Häussler, J., & Juzek, T. (2021). Variation in participants and stimuli in acceptability experiments. In G. Goodall (Ed.), *The Cambridge handbook of experimental syntax* (pp. 97–117). Cambridge University Press.
<https://doi.org/10.1017/9781108569620.005>
- Koo, T., & Li, M. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, 15(2), 155–163.
<https://doi.org/10.1016%2Fj.jcm.2016.02.012>
- Kobzeva, A., Sant, C., Robbins, P., Vos, M., Lohndal, T., Kush, D. (2022). Comparing island effects for different dependency types in Norwegian. *Languages*, 7(3), Article 97.
<https://doi.org/10.3390/languages7030197>
- Kush, D., Lohndal, T., & Sprouse, J. (2018). Investigating variation in island effects: A case study of Norwegian wh-extraction. *Natural Language & Linguistic Theory*, 36(3), 743–779.
<https://doi.org/10.1007/s11049-017-9390-z>
- Kush, D., Lohndal, T., & Sprouse, J. (2019). On the island sensitivity of topicalization in Norwegian: An experimental investigation. *Language*, 95(3), 393–420. <https://doi.org/10.1353/lan.2019.0051>
- Langsford, S., Perfors, A., Hendrickson, A., Kennedy, L., & Navarro, D. (2018). Quantifying sentence acceptability measures: Reliability, bias, and variability. *Glossa: A journal of general linguistics* 3, Article 37.
<https://doi.org/10.5334/gigl.396>
- Laurinavichyute, A., Himanshu, Y., & Vasisht, S. (2022). Share the code, not just the data: A case study of the reproducibility of articles published in the Journal of Memory and Language under the open data policy. *Journal of Memory and Language*, 125, Article 104332. <https://doi.org/10.1016/j.jml.2022.104332>
- Loken, E., & Gelman, A. (2017). Measurement error and the replication crisis: The assumption that measurement error always reduces effect sizes is false. *Science*, 355(6325), 584–585.
<https://doi.org/10.1126/science.aal3618>
- Marty, P., Chemla, E., & Sprouse, J. (2020). The effect of three basic task features on the sensitivity of acceptability judgment tasks. *Glossa: A journal of general linguistics*, 5, Article 72.
<https://doi.org/10.5334/gigl.980>

- Maxwell, S., & Delaney, H. (2004). *Designing experiments and analyzing data: A model comparison perspective* (2nd ed.). Psychology Press. <http://doi.org/10.4324/9781410609243>
- Moore, C. (2016). *gtheory: Apply Generalizability Theory with R*. R package version 0.1.2. Retrieved from <https://CRAN.R-project.org/package=gtheory>
- O'Brien, R. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality and Quantity*, 41(5), 673–690. <https://doi.org/10.1007/s11135-006-9018-6>
- Pañeda, C., Lago, S., Vares, E., Veríssimo, J., & Felser, C. (2020). Island effects in Spanish comprehension. *Glossa: A journal of general linguistics*, 5, Article 21. <https://doi.org/10.5334/gjgl.1058>
- Pañeda, C., & Kush, D. (2021). Spanish embedded question island effects revisited: An experimental study. *Linguistics*, 60(2), 463–504. <https://doi.org/10.1515/ling-2020-0110>
- Peer, E., Rothschild, D., Gordon, A., Evernden, Z., & Damer, E. (2021). Data quality of platforms and panels for online behavioral research. *Behavior Research Methods*, 54(4), 1643–1662. <https://doi.org/10.3758/s13428-021-01694-3>
- R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Rajaratnam, N. (1960). Reliability formulas for independent decision data when reliability data are matched. *Psychometrika*, 25(3), 261–271. <https://doi.org/10.1007/BF02289730>
- Rietveld, T., & van Hout, R. (1993). *Statistical techniques for the study of language and language behaviour*. Mouton de Gruyter. <https://doi.org/10.1515/9783110871609.fm>
- Rodríguez, A., & Goodall, G. (2023). *Clitic left dislocation in Spanish: Island sensitivity without gaps* [Conference presentation]. Annual Conference on Human Sentence Processing 36, Pittsburgh, PA, United States.
- Roettger, T. (2021). Preregistration in experimental linguistics: applications, challenges, and limitations. *Linguistics*, 59(5), 1227–1249. <https://doi.org/10.1515/ling-2019-0048>
- Rouder, J., & Haaf, J. (2018). Power, dominance, and constraint: A note on the appeal of different design traditions. *Advances in Methods and Practices in Psychological Science*, 1(1), 19–26. <https://doi.org/10.1177/2515245917745058>
- Schoenmakers, G. (2023). Linguistic judgments in 3D: The aesthetic quality, linguistic acceptability, and surface probability of stigmatized and non-stigmatized variation. *Linguistics*, 61(3), 779–824. <https://doi.org/10.1515/ling-2021-0179>
- Schoenmakers, G., & van Hout, R. (2024). The internal consistency of acceptability judgments from naïve native speakers. *Nederlandse Taalkunde*, 29(1), 82–88. <https://doi.org/10.5117/NEDTAA2024.1.007.SCHO>
- Schoenmakers, G., & Stoica, I. (2024). An experimental investigation of *wh*-dependencies in four island types in Romanian. *Glossa: A journal of general linguistics*, 9. <https://doi.org/10.16995/glossa.15193>
- Schütze, C. (2016). *The empirical base of linguistics: Grammaticality judgments and linguistic methodology*. Language Science Press. https://doi.org/10.26530/OAPEN_603356 (Original work published 1996 by University of Chicago Press).
- Schütze, C., & Sprouse, J. (2014). Judgment data. In Podesva, R., & Sharma, D. (Eds.), *Research methods in linguistics* (pp. 27–50). Cambridge University Press. <https://doi.org/10.1017/CBO9781139013734.004>
- Shavelson, R., & Webb, N. (1991). *Generalizability Theory: A primer*. SAGE.
- Shrout, P., & Fleiss, J. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2), 420–428. <https://doi.org/10.1037/0033-2909.86.2.420>

- Shrout, P., & Rodgers, J. (2018). Psychology, science, and knowledge construction: Broadening perspectives from the replication crisis. *Annual Review of Psychology*, 69, 487–510. <https://doi.org/10.1146/annurev-psych-122216-011845>
- Sönning, L., & Werner, V. (2021). The replication crisis, scientific revolutions, and linguistics. *Linguistics*, 59(5), 1179–1206. <https://doi.org/10.1515/ling-2019-0045>
- Sprouse, J. (2007). *A program for experimental syntax: Finding the relationship between acceptability and grammatical knowledge*. Ph.D. dissertation at University of Maryland.
- Sprouse, J. (2018). *Experimental syntax: Design, analysis, and application*. Lecture slides, last updated on 14 December, 2018, available at <https://www.jonsprouse.com/courses/experimental-syntax/slides/full.slides.pdf>
- Sprouse, J., & Almeida, D. (2013). The empirical status of data in syntax: A reply to Gibson and Fedorenko. *Language and Cognitive Processes*, 28(3), 1–7. <https://doi.org/10.1080/01690965.2012.703782>
- Sprouse, J., & Almeida, D. (2017). Design sensitivity and statistical power in acceptability judgment experiments. *Glossa: A journal of general linguistics*, 2, Article 14. <https://doi.org/10.5334/gjgl.236>
- Sprouse, J., Caponigro, I., Greco, C., & Cecchetto, C. (2016). Experimental syntax and the variation of island effects in English and Italian. *Natural Language & Linguistic Theory*, 34(1), 307–344. <https://doi.org/10.1007/s11049-015-9286-8>
- Sprouse, J., & Villata, S. (2021). Island effects. In Goodall, G. (Ed.), *The Cambridge handbook of experimental syntax* (pp. 227–258). Cambridge University Press. <https://doi.org/10.1017/9781108569620.010>
- Vasishth, S., & Gelman, A. (2021). How to embrace variation and accept uncertainty in linguistic and psycholinguistic data analysis. *Linguistics*, 59(5), 1311–1342. <https://doi.org/10.1515/ling-2019-0051>
- Winter, B. (2019). *Statistics for linguists: An introduction using R*. Routledge. <https://doi.org/10.4324/9781315165547>
- Zimmerman, D., & Zumbo, B. (2015). Resolving the issue of how reliability is related to statistical power: Adhering to mathematical definitions. *Journal of Modern Applied Statistical Methods*, 14(2), 9–26. <https://doi.org/10.22237/jmasm/1446350640>